

DEF CON 31 - Second Breakfast Implicit & Mutation Based Serialization Vulns in NET - Jonathan Birch

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Description

Exploits of insecure serialization leading to remote code execution have been a common attack against .NET applications for some time. But it's generally assumed that exploiting serialization requires that an application directly uses a serializer and that it unsafely reads data that an attacker can tamper with. This talk demonstrates attacks that violate both of these assumptions. This includes serialization exploits of platforms that don't use well-known .NET serializers and methods to exploit deserialization even when the serialized data cannot be tampered with. Remote code execution vulnerabilities in MongoDB, LiteDB, ServiceStack.Redis, RavenDB, MartenDB, JSON.Net and the .NET JavaScriptSerializer are all

demonstrated. Techniques to both scan for and mitigate these vulnerabilities are also discussed, along with methods and obstacles for exploiting serialization in .NET 6+.

Transcript

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So this is Second Breakfast, Implicit and Mutation-Based Serialization Vulnerabilities in .NET. I'm Jonathan Birch, I hack Office for Microsoft. And if you want to talk to me about this presentation- [inaudible 00:00:16]. ... Oh. I'm Jonathan Birch, I hack Office for Microsoft. If you want to talk to me about this presentation or anything else security related, please contact me on Mastodon. In this talk, I'm going to explain remote code execution vulnerabilities I found in several .NET NoSQL engines. I'll talk about what I'm calling a Mutation-based Serialization Vulnerability. I'll explain how you can bypass SerializationBinders, and how you can defend against all of that. Also, some bonus stuff just for DEFCON at the end. So a lot of my day job is doing security design reviews. I have a bunch of meetings every week, where I'm looking at data flow diagrams, telling people do it this way instead, that kind of thing. And something I do in those meetings anymore to add value is I'll just open up the repo that I've never seen before and spend a half hour while I'm talking to them, figuring out what are the actual vulnerabilities here, so I can talk to them about that too. So the research that led to this talk happened because last fall, I saw an article on a certain orange-colored website about LiteDB, and it's a .NET open source library for a NoSQL engine. I thought, okay, I'll just go do a security review of their repo. And so when I did that, I saw this code. And what jumps out to me in this code specifically, is this line, this call to `Type.GetType` that's being passed a string. And to me at least, that's a really bad sign. But why is that a bad sign? So `Type.GetType` in .NET is used to convert a string that's like the name of a .NET type into an actual type object. And what that means is that someone's going to be giving you the name of a type generally for an object you're then going to create. So they're telling you what kinds of objects you're creating, which means they're telling you essentially what code you're going to run. And that means that if someone malicious comes along and they give you a malicious type name and you're running malicious code. So how does that show up in LiteDB? Well, LiteDB is a NoSQL engine, and generally when an application is using it, it's just saying to LiteDB, store this object I've got, and maybe at this index with these properties, that kind of thing. What LiteDB is doing, is it's going to put that in a file, so it serializes it with a serializer called BSON Mapper into a BSON stream and stores that BSON stream in a file. And when the application wants the object back, it de-serializes that BSON back into an object again. So BSON is just a binary encoding of JSON. So you can take that file LiteDB makes decode it back into JSON, and if you do that, it looks about like this. And if you look at this JSON here, it's got this underscore type key in it and after that it's got the name of an actual .NET type. So what LiteDB is doing with `Type.GetType` is it's taking that type name, it's passing it to `Type.GetType` to actually create the object during that de-serialization. So the data in that BSON is telling LiteDB what types it's going to make. And that means that this is just a polymorphic serializer behind the scenes and that means it's

got all of the classic vulnerabilities that polymorphic serializers have. But let's explain a little bit what I'm talking about there. So .NET serialization vulnerabilities. .NET serialization vulnerabilities generally happen because you have a polymorphic serializer, a serializer where the data is telling you what type it wants to be. So in this case we've got a box, it's themselves describing data, it says it's a cat, it's labeled cat. When your serializer sees that label, it says, well I guess I'm making a cat object and it just makes a cat object and then sets some properties from the contents of that box. And the problem with this is someone malicious can come along and give you a box labeled pain. Your serializer says, well I guess I'm making some pain then and you're having a bad time. So if you want a little more information in general on how .NET serialization works as far for those sorts of serialization exploits, I suggest looking at the Friday the 13th JSON talks from Black Hat 17, but I'll also give you a few examples of what this pain is here. So in general, what I'm talking about pain here, what I'm talking about is types that where if you tell an application to create them, they're going to have side effects that can further exploit goals. Either when you call their constructor set their properties, maybe the destructor will do something that you want to have happen. Here's a couple of classic ones. Assembly and solar has a path property. When you set the path property, it just calls load library on whatever path you give it. So if you've been able to drop a malicious DLL on a server somewhere, you're going to be able to do DLL injection. It turns out that if you give it an HTTP URL, it will actually go and request the file from that URL. It won't load it, but this is still really useful for getting ping backs if you're attacking a server somewhere. Object data provider is sort of the more classic RCE gadget for .NET serialization. It's got properties where if you set the properties on it, you can call any static method on any type in .NET with whatever string and integer parameters you want. So people usually call Process.Start just to pivot to shell injection. Okay. So that's how .NET serialization vulnerabilities work. Now I'll go back to LiteDB and explain what I'm calling implicit serialization vulnerabilities. So here's a remote code execution exploit for LiteDB. If you look at this JSON Blob here, this is going to be the JSON, we're going to put into LiteDB. It's got that underscore type key where it's telling you what type it wants to be. It's going to be an object data provider that's that gadget that lets you call any static method you want. We're going to call Process.Start and launch calc, and all we're going to do with this JSON is convert it to BSON and then pass it into BSON mapper to de-serialize it. It's worth noting that when you call the BSON mapper de-serialized method, you tell it what type you expect and it does not matter. So LiteDB will actually respect the type the data says it wants to be. It'll go make it. It'll set its properties, your serialization exploit gadget will run. It'll then try to do a cast that will fail and probably throw an exception, but it doesn't matter, your code is already run. So going back to how LiteDB works, if you remember LiteDB uses BSON Mapper to read its own storage. So the problem here is that anything that can put malicious data in the database file can achieve RCE. It's also worth noting that when a client's using LiteDB, they don't know they're using a serializer probably. They're almost never going to call BSON Mapper directly. So this is an implicit serialization vulnerability. The client is vulnerable to the serialization that's in LiteDB even though they're not using the serializer LiteDB is. And this makes that client vulnerable in a lot of ways. First of all, LiteDB, when it stores a database file, it just stores a local file. So a lot

of applications use LiteDB just to write document formats or save files for games. So if you provide someone a document or a file that they're reading with LiteDB, you can put an RCE payload in there and it'll run code when they open it. LiteDB is a NoSQL engine. It also has queries, which means you can achieve query injection and then you can put a malicious record in the database and when someone reads that record, you've got RCE and there's one extra special exploit technique I figured out for this that I'll come back to. So what do you do when you find a remote code execution vulnerability in a major open source library? Well, you look at all the other libraries that are like it and find the same vulnerability there. So here's essentially the same vulnerability in MongoDB. So here's JSON payload. It's got this underscore TKey that's the type specifier Mongo uses, that's the way the data says what type it wants to be. We're again making an object data provider. We're going to then again call Process.Start and launch calc. And again, all we're going to do with this is convert it into BSON and then de-serialize it with the BSON serializer that MongoDB uses. Now it's worth noting that you also tell the BSON serializer what type you expect and that does matter. So in MongoDB, it will actually check that the type the data says it wants to be is assignable to the type it's expecting to get before it creates it. So if you want to exploit MongoDB, you have to find something in the data structure the application's expecting that's open-ended so you can fit a serialization gadget into it. That said, it's almost always there when I've looked at applications that use databases, there's almost always either an interface member on something or just a member of type object is like a catchall. And so you just put your serialization gadget in that slot and you still have RCE. Because MongoDB again uses this BSON serializer to read its own database. It means that anything that puts malicious data in the database gets your remote code execution. MongoDB uses a client server architecture most of the time. So you can have one malicious client that talks to the database, puts a bad record in there, some other client reads that you're running code on that other client. You can also do again, query injection. I will say that I was only able to exploit the .NET driver from MongoDB, not Java or Python. I won't say you can't, just that I couldn't figure out how to do it. And then we're going to move on to yet another remote code execution vulnerability for yet another NoSQL engine. This one is RavenDB. So once again, we've got a JSON payload. This one, the type it says it wants to be is a Windows identity. Windows identity is a JSON.NET or RCE serialization gadget. I suggest looking at ysoserial for details of how it works, but essentially unless you pivot to a binary format or payload, which you then use to get RCE, and in this case what we're going to do with this payload is we're just going to actually make a connection to the database server and then do a put request to stick the JS in the database. So RavenDB lets you interact with the database in two ways. You can read or write JSON to it or you can read or write objects from it. And so what you can do is if you put malicious JSON in the database and it's read as an object that gets you RCE. Now it's worth noting here again that the type that's being expected matters. There's an assignability check like there was in MongoDB, and so you have to actually have some way to fit your payload into the expected types. This again is an implicit serialization vulnerability because RavenDB is just using JSON.NET with unsafe settings to read its own storage. So anything that puts bad data in the database gets you RCE and that special attack pattern that I mentioned for LiteDB works here also. And I looked at some things

that were NoSQL but not strictly databases. So here is a remote code execution vulnerability for ServiceStack.Redis. ServiceStack.Redis is sort of a NoSQL wrapper for a Redis cache. So here we've got some JSON. It says what type it wants to be. In this case it's saying it's going to be an assembly installer. That's one that lets you call load library on an arbitrary path. We're just going to say I dropped a malicious DLL somewhere for it to read. And then what we're going to do with this JSON, is we're going to stick it in the Redis cache and then read it back. So when we stick to JSON in the reddish cache as JSON, it's just JSON, but when you read it back as an object using ServiceStack.Redis, it's going to turn into whatever object we say it wants to be and it can run code. Now again, the type matters. So when you're using ServiceStack.Redis, you tell it what type you're expecting and the payload you're using has to fit into the object draft you're going to have there. This is again an implicit serialization vulnerability. So because ServiceStack.Redis is using unsafe serializer to read that Redis cache, anything that lets you put bad data in the Redis cache also gets you RCE. And that extra pattern I keep talking about works here also. And now I'm going to tell you a little bit about how I discovered it and that's because I was looking at one other NoSQL engine in .NET and that's MartenDB. So MartenDB like RavenDB uses JSON.NET with unsafe settings to read its own storage. But unlike RavenDB, it doesn't just let you write JSON. You can only read and write objects from the database. And when I tried to write an object to MartenDB that was just like a serialization exploit gadget, it would choke on it every time I tried. So I played with that a lot and I found a completely different way to attack it and that's what I'm calling a serialization mutation attack. So here's a remote code execution payload for MartenDB. This one's a little bit different. So what I'm going to do here is I'm going to make in .NET a dictionary of string, string. This could be a hash table two, but you need something that's got key value pairs mapping a string to a string. And what I'm going to put in that dictionary is first a dollar type key paired with the name of .NET type. In this case workflow designer, which is another JSON.NET serialization gadget unless you do an arbitrary SAML read. Again, look at ysoserial for details about how it works. And then I'm going to do one other thing where I'm going to set another key value pair where this is the name of a property on workflow designer, which is the one you set the SAML with to get an arbitrary SAML parse out of it. And what I'm going to do with that is I'm just going to write it to the database. So if you write this object to the database and then read it back, that will run code, it will launch calc, and that's kind of weird, we're writing a dictionary to the database, but when you read it back, it runs calc. Why does that work? So this is using JSON.NET as its serializer. So let's look at how this works in JSON.NET. So in JSON.NET, if you serialize one of these dictionary string, strings, like, say you have a dictionary where you've got a key named fruit and the value pair, it uses the key names exactly as you provide them. So your key as fruit is just going to go into the serialized JSON as the key named fruit. Now if you compare that to what an RCE payload looks like for JSON.NET, here's an example of one. You've got that type specif hierarchy, that dollar type, and then you've got a .NET type name as a value for it and so on. And the key trick with this is there's nothing special about that dollar type key. There's nothing special about the type specifier used really in any of the serializers I've talked about. So you can make a key for a string, string dictionary named dollar type, and it turns out that if you serialize one of those string,

string dictionaries in JSON.NET or most of the other serializers I talked about, it just uses the key directly. It doesn't filter it out even though it matches its type specifier. So what happens there is that you're essentially colliding with the syntax of what that RCE payload looked like. You're making it just output data where it's going to be the type specifier. So when this object gets serialized and then de-serialized, it mutates into a different type entirely and you can choose whatever type you want. And this doesn't just work for dictionary, it works for hash table, JObject, ExpandoObject and like 50 others. Just go look around the .NET for things that have key value structures. And because of this mutation attack, you can do a lot of attacks on serialization that usually weren't considered possible. So say you have a website that takes some key value data and this is a screenshot from a certain website you might recognize that did not have this vulnerability but does let you enter key value data where you can choose both the keys and the values and you send that to a web front end with a form post and on the web front end, they're going to make a dictionary or a hash table with the data you sent them and later on say that dictionary or hash table gets serialized through a database or a cache and then later on after that, it gets read back from that database or cache using an unsafe serializer. Well, in that case you can provide key value data where you're using that type specifier is the key name and when it gets read from that cache on some backend server, you can make it so it mutates into whatever object you want. It's now your serialization exploit gadget, it's now that pain and you've got remote code execution whenever that happens. So I was saying that was that extra attack pattern against all of these NoSQL engines. Here's how that works against LiteDB. So LiteDB, you can make a dictionary string, string. You put your payload in there, so you've got that underscore type key specified with a type name, in this case, assembly installer. So we're going to do again an arbitrary called a load library. We're saying we're loading a malicious DLL and all you do is you write it to the database and read it back. Now this hack is kind of magic. It doesn't matter where this is in the data structure. If you can control two key value pairs in some part of some object graph, if it gets saved and read back, that gets you RCE. And this actually makes a lot of attacks really ridiculous and easy. Like there's some application that's just asking you for a couple of string pairs and later you get RCE when it gets read from the database. And this kind of attack works with a lot of serializers. Here's a list of all the ones I know, these are all the .NET serializers I know where this sort of mutation attack works. You'll notice these are all JSON and BSON serializers. This doesn't work for binary formatter because it doesn't use anything like this kind of syntax. And it's also worth noting that this doesn't work against MongoDB serializer. You'll note that there's different requirements for this to work in certain cases. Some of them you have to control the first key value pair. Some of them there's that assignability check. Some of them there's other specific unsafe properties you've got to set. So there's a lot of variability in how exploitable mutation can be. For example, JSON.NET like most popular Open Source library in the world as far as I'm aware, you can get mutation attacks against it, but it's a little bit tricky. You have to control the first key value pair. The only one it'll read the type specifier from. It has to be serialized using unsafe settings. It has to be serialized in such a way that it doesn't double up the type specifier. And in general, I would suggest experimenting with this. There's a lot of weird special cases and nuances, but sometimes you can make this work and you can

use this to exploit things where the serialization is just not something you can touch. The serialization happens on the backend, you can't see the de-serialized data. Maybe it even gets an HMax stuck on it. It doesn't matter. On the other hand, there is serializer where this just works all the time. So if you've got JavaScript serializer with a type resolver, it will go out of its way to make the exploit work. It doesn't matter if you've got other entries in your key value storage. It doesn't matter if your type specifier is the last thing in the list. It doesn't matter if the assignability doesn't make sense, it'll just run the code and it'll just work. You can also do a blind scanning technique to find these vulnerabilities. If you've got websites out there that you think are running on .NET and might have some backend cache or database, you can just stick key value data in there. You can use that assembly installer ping back trick I've talked about and you can wait to see if you later get a request back to your own server so you can stick data in services blind, use, maybe, unique URLs for your ping back so you can see which ones to talk back to you and then maybe later on when it gets read from the cache, you'll get a request, and you know there's an RCE happening there. You can defend and get these attacks in a bunch of ways. If you're writing an application, the best thing to do is don't use a polymorphic serializer, and I'm going to say that a lot in this talk. Don't use a serializer where the data gets to tell you what type it wants to be. If you have to, for some reason, you can try using a SerializationBinder, but I'm about to tell you how to bypass SerializationBinder is so grain of salt there. So here's how to bypass SerializationBinders. In general, if you're talking about using a SerializationBinder for security purposes is acting like a filter. What's happening is during the serialization, your binder is going to get asked about every type that's directly referenced in the serialized data and it'll say, okay, this is a type I expect, go ahead. Or if you're doing it right, you throw an exception when some type you don't expect shows up. And that means if someone throws pain at you, you say, I don't know what pain is, you throw an exception and nothing bad happens. So if you're doing it right, your SerializationBinder looks sort of like this. What this is just an allow list of all the expected types. It's a string allow list for both the assembly name and the type from the assembly name. And if you see one that you don't expect, you throw an exception. The problem is a lot of times people don't do this. Usually what you see for a SerializationBinder is something more like this where you have some assembly you've written yourself. You say, well, I trust all my types. There's no crazy serialization gadgets in there. So if the type says it's coming from my assembly, I'll just allow it and I'll use Type.GetType to actually get the type out of there. And the problem with this approach is, say, your assembly contains a type like this. This is an initialized list type we're going to invent that just derives inherits from the just generic list type in .NET and we're adding another property to it. That seems pretty harmless, but if you've got a type like this in that assembly that you're trusting, you can exploit it like this. So here's JSON payload to bypass that SerializationBinder I just showed you. The type it's saying it wants to be is initialized list the type I just mentioned, but the type in the template for that is assembly installer. That one that lets you do an arbitrary load library. The trick with this is initialized list is the type you're getting asked about that comes from your assembly so that binder is okay with it. And you'll notice that assembly installer is only ever referenced inside the template here, but when this values part of the JSON Blob gets de-serialized, it'll still make

an assembly installer object. It'll still set its path property and you're still doing load library. So your RCE is still working. There's a trick here that when you serialize this with JSON.NET, it'll put another dollar type entry in there for the assembly installer, but you don't need it. It can figure out what it's supposed to be if it's not there. So you can just take it back out again and then bypass the binder. There's another trick you can use to bypass SerializationBinders that I'm calling contagion. So for JSON.NET and a lot of other BSON and JSON serializers, you've got that dollar type entry that's saying what type it's supposed to be, but that object you're creating has members, properties, constructor arguments that have their own types and you don't have to specify the types of those in the JSON. If you serialize it will, but if you don't, the serializer will just figure it out. So if there's some serialization gadget you want to use and it's a member of something that's a member of something, that's a member of something that you can get past the binder, you can use that one object that has it in its tree somewhere to get it past the binder. Here's one example. So Windows identity that I mentioned before is a JSON.NET RCE gadget. It lets you pivot to binary formatter and you could have a SerializationBinder that specifically block lists that. And if you do that you can still get it by. The way you do that is you use a WindowsAuthenticationEventArgs object. So if you serialize WindowsAuthenticationEventArgs, it has a constructor argument called identity whose type is Windows identity. When you serialize that, the only type going by the binder is this dollar type because the only things that the binder gets asked about are the types explicitly listed in the JSON, but when the identity part of it gets de-serialized here, it'll still make a Windows identity object. It'll still set its properties. You still get RCE. So that's a lot of problems. What if this has been fixed? So I am with Microsoft, I practice responsible disclosure. I told all of these people about these issues in January or earlier. Everybody except MartenDB made some sort of fix. MartenDB as far as I've checked last, still hasn't. MongoDB has fixed it pretty solid. I don't know how to bypass it. They added a pretty strict allow list on the types that can get created during the serialization. It seems good. Everybody else here added a block list. LiteDB also added an assignability check so that puts some about on par from MongoDB before they made a fix. I just showed you how to bypass block lists. I can bypass all of the fixes for LiteDB, RavenDB in ServiceStack.Redis. Like I said, MartenDB didn't get fixed. I told all of these people, "Look, I can still bypass these fixes." They said we're not making additional fixes. So there's remote code execution vulnerabilities that are live right now for all of those libraries worth looking at. The mutation attack I mentioned against JSON.NET is not being fixed. I talked to James Newton-King, the maintainer for JSON.NET about this. He said essentially it's supposed to do that. So that behavior is not going to be changed. If you've got some application that's using JSON.NET, it might be vulnerable to mutation. Similarly, JavaScript serializer, the .NET serializer I mentioned that works really well for mutation attacks will not be fixed. I talked to the .NET team about this. They said, we tell people don't use JavaScript serializer. We tell people don't use type resolver, so we're not going to fix them, just don't use them. So best practice is to avoid these issues. First thing I keep saying, don't use a polymorphic serializer, don't roll your own. It's a really bad idea. Don't have that data let you tell you what type it wants to be. It leads to all kinds of problems. Don't call Type.GetType because if you're doing that, you're probably writing a polymorphic serializer

even if you don't realize you're doing it. Similarly, in Java, don't call class for name. Also remember that you can't use an unsafe serializer anywhere because mutation attacks. If it's on your backend, if you've got an hvac, it doesn't matter for mutation. You can attack, like deep copy tricks. A lot of people use serializers to just deep copy objects. You can get RCE out of that because of mutation, which is silly. So don't use unsafe serializers anywhere. This one hurts, but maybe don't read untrusted data from NoSQL. Mongo might be okay, but it looks like every other NoSQL engine I've looked at. You can get RCE if someone can write bad data and you can read it back. Now I only really looked at .NET NoSQL engines, but there's kind of a pattern here with NoSQL where they say, just take this object and then later on you can tell the NoSQL engine, I want the object back and I don't care what it is. Anytime you have that pattern, there's a polymorphic serializer in there somewhere and you can probably get RCE out of it if you want a research project or want to go find libraries that are vulnerable, maybe you look at Java or Python NoSQL engines. Don't try to use SerializationBinders. It's possible to write a good one, but nobody ever does it. Everybody screws this up. I see bypasses for these all the time. In general, you're better off architecting your application so you don't need a SerializationBinder in the first place. And now I'm going to get into the bonus stuff that's just for DEFCON. So I've been talking about .NET framework exploits, but what about .NET 6 plus? Everyone seems to think that's different. It's not going to have these serialization vulnerabilities. So since 2016, .NET has had a couple of different flavors. There's .NET framework and then there's .NET, that's what they want me to call it. But .NET 6 through 8, those are the ones people use right now. All these exploits have been .NET framework exploits. So the ones that are in ysoserial, but what about .NET 6 plus? Can you do this sort of stuff? There is this myth that .NET 6 plus is immune to serialization attacks. So here's the CVE Mongo opened for the vulnerability I found, and it's an eye chart there, but I'll underline the important part here and what they said was it must be running Windows hosts using full .NET framework not .NET Core. That's not true. Similarly, if you look at ysoserial, they have a bug there, which is, is there any gadget for .NET Core? And no one has found one there. So admittedly exploiting .NET 6 plus is harder. A lot of the exploitable types, the gadgets that people use in .NET framework just aren't there in the framework at all. There's also this whole issue where .NET, .NET 6 plus whatever we're calling it, actually divides the framework into three parts. There's .NET Core. There's ASP.NET, and there's the Windows desktop part. And depending on how you build your application, you only get some of the DLLs, they're even visible at all. So you can only use gadgets from the ones that are appropriate to how you built your application. So the ASP.NET Core app types and the Windows desktop app types won't be found by Type.GetType. You can't de-serialize them unless you've built with an ASP.NET app or a Windows desktop app, but you can still do de-serialization exploits. So object data provider is still there in .NET 6, it's still there in .NET 8, you have to have Windows desktop loaded for it to work. But if it's loaded, the payloads I've been showing you for object data provider, they just work fine. There are also some serialization gadgets that do exist in .NET Core app. And now I'm going to go to the parts that aren't on my slides because I think I've got some time left. So I told you earlier how you can get a painting back with assembly installer. There's a similar trick you can do in .NET 6, in .NET 8 and so on. And there's a few types you can use to do that. One

of them is assembly dependency resolver. Another one is system.io.file info. The trip with both of those is they will call create file with whatever string you give them. So assembly dependency resolver has a constructor argument which is just a path, and it'll just call create file on that path. The file info type, you can set the read only flag on a file with it and you can do that, however. So you can also use that as a payload to do DOS because you can make files on the server read only, but when you do that, it'll also call create file on that path. Now there's a thing with create file on Windows where if you give it an internet SMB URL or a WebDAV URL, it'll go talk to that. It's happy to do it. So if you've got an SMB server on the internet or a WebDAV server somewhere, you can use those gadgets to get it ping back to that server. Now a lot of times if they're doing their OPSEC correctly, SMB outbound will be blocked on the server you're trying to attack. And sometimes the WebDAV redirector won't exist in the Windows install. So it's a little bit iffy, but in my experience, usually one or the other of those works. So you can still get a ping back essentially by using assembly dependency resolver or file info, setting the read only flag, and then giving either an internet SMB URL or a WebDAV URL. And that's actually all I had. So are there any questions? Oh, thanks. I saw one over here. Sorry, can you tell me that one more time, sorry? You mentioned some games [inaudible 00:30:43]? Okay, the question was do I have any examples of the vulnerable games? I don't remember which ones they are, but what you can do is go on GitHub or if you go on to NuGet... Okay, so two approaches, go to NuGet, look at our projects that are dependent on LiteDB and then look at where those get used. Or just search on GitHub for uses of LiteDB for things that look like games. That's how I found a bunch of our examples of that before. Yes, over here. [inaudible 00:31:19] Sorry, you said that after you're reading the object, it will actually initialize the object injection, right? So how did you find that in the first time? Because it's not something that you typically think about when you start this research. Okay, to clarify with the question, how did I realize that the vulnerability was there in the code that I was looking at to begin with? Yeah, because you said I can control the key, I can control the value and then when I read it will initialize the object injection. Right? Oh, okay. So as far as that goes, if you're looking for polymorphic serializers where people are rolling their own, which is a really bad idea, in general, what I look for when I'm doing these with code reviews is calls to Type.GetType that are just being passed a string. Especially if the string somehow comes from user data. Type.GetType being called on a user provided string almost always is either unsafe serialization or someone doing really sketchy with reflection and evokes. And both of those are probably RCE. Does that explain it? Yes it is. Thank you. Okay. In the middle here. AWS [inaudible 00:32:46]. Okay, the question was did I look at the AWS [inaudible 00:32:53]? DynamoDB [inaudible 00:32:54]? I did at least... Okay. The question was did I look at DynamoDB NoSQL? I glanced at it. I did not do an in-depth test. It did not have as easy of an exploit as the other ones I looked at. I won't say it's safe. And then there was one over here. Okay, the question here was, the assembly installer exploit that I'm loading malicious DLL with and how can I make sure the malicious DLL is there present on the system? In that case, the assumption is you've got some other exploit that lets you do a file drop. Something I should explain is the cases where I was using assembly installer exploits, that's because the string is shorter than the object data provider ones. Object data provider works for all of these two. So do lots of other .NET

Gotchas. Assembly installer, I feel like honestly is more useful for doing a ping back to an HTTP URL because that lets you do that blind scanning I mentioned. And one, way in the back. You might have to come forward for me to hear you. So the question was whether Automapper and fast mapper that also used that sort of reflection under the covers are also vulnerable. I'll admit I haven't looked at those. I'm honestly not quite sure what you're referring to. So maybe we can talk a little bit more about it after this. And well, if there's no other questions, I'll finish there. Thank you.

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